



BAULKHAM HILLS HIGH SCHOOL

Assessment Task 2 2013
YEAR 11

Mathematics Extension 1

General Instructions

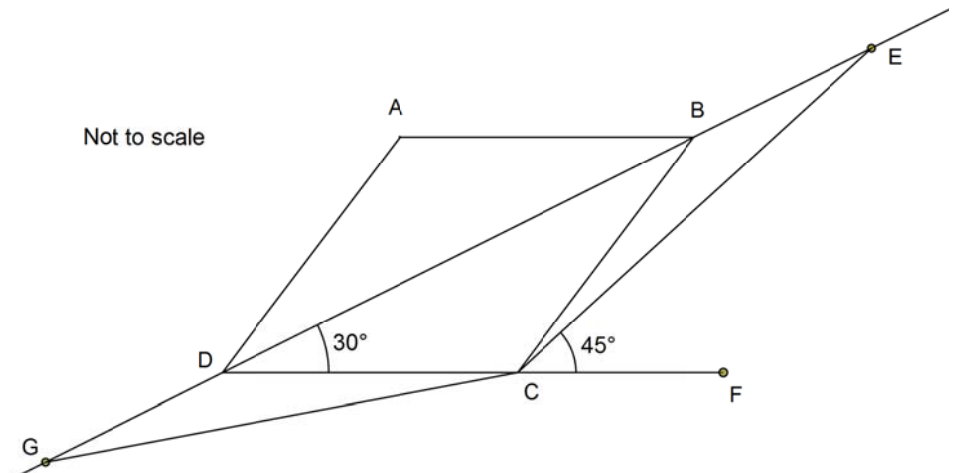
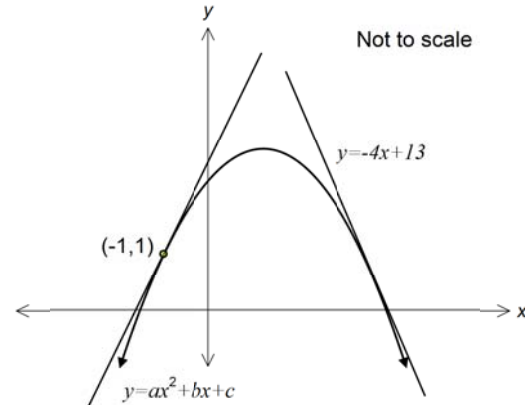
- Reading time – 5 minutes
- Working time – 1 hour and 10 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 1-14
- Marks may be deducted for careless or badly arranged work

Total marks – 48

Exam consists of 2 pages.

This paper consists of 14 Questions.

1.	If A and B have coordinates $(1, -2)$ and $(-3, 4)$ respectively, find the point $P(x, y)$ which divides the interval AB externally in the ratio 2: 3	2
2.	Without solving simultaneously, find the number of times $y = x - 4$ cuts the circle $(x - 1)^2 + (y - 2)^2 = 18$	3
3.	Find the exact value of $\sin 105^\circ$	3
4.	Find the acute angle between the lines $y = \frac{x}{3} - 2$ and $2x + y + 1 = 0$ to the nearest degree	3
5.	(i) Prove $\frac{\sin 2x}{1 - \cos 2x} = \cot x$ (ii) Hence find the exact value of $\cot 15^\circ$	2 2
6.	Show $\frac{d}{dx} \left(\frac{(2x-1)^3}{\sqrt{2x-1}} \right) = 5\sqrt{(2x-1)^3}$	2
7.	Solve $2 \sin \theta - \cos \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$ using the t -method where $t = \tan\left(\frac{\theta}{2}\right)$	4
8.	(i) Express $\sqrt{3} \sin \theta + \cos \theta$ in the form $A \sin(\theta + \alpha)$ (ii) Hence solve $\sqrt{3} \sin \theta + \cos \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$	2 2
9.	Differentiate $f(x) = \frac{2}{x^2}$ from first principles.	3
10.	Solve for $0^\circ \leq x \leq 360^\circ$ $2 \sin 2x = \cos x$	3

11.	Not to scale  $ABCD$ is a rhombus. DB is produced to E and BD is produced to G. $\angle BDC = 30^\circ$ and $\angle ECF = 45^\circ$ (i) Prove $\triangle BCE$ is isosceles (ii) If $DG = 2BD$ what is the ratio of the area of $\triangle BGC$ to the area of the rhombus $ABCD$?	4 2
12.	Find the point(s) on the curve $y = \sqrt{1 - x^2}$ where the normal is inclined at 135° to the positive direction of the x-axis.	4
13.	Solve for $0^\circ \leq x \leq 360^\circ$ $\frac{6^{\sin x} + 3^{\sin x}}{2^{1+\sin x} + 2} = \frac{\sqrt[3]{3}}{2}$	3
14.	Not to scale  In the diagram above, the tangent at the point $(-1, 1)$ on the parabola $y = ax^2 + bx + c$ has a gradient of 4. The line $y = -4x + 13$ is also a tangent to the parabola. Find the values of a, b and c.	4
End of Exam		

Solutions. / 48

1. $A(1, -2)$ $B(-3, 4)$

[2] $-2:3$

$P\left(\frac{-2(-3)+3(1)}{-2+3}, \frac{-2(4)+3(-2)}{-2+3}\right)$

$P(9, -14)$

Award 1 for internal division (3,2)

2. [3] If $y = x - 4$ cuts circle at 2 places then the distance from $(1, 2)$ to the line must be less than the radius of the circle i.e. $\sqrt{18}$.

$x - y - 4 = 0$ $(1, 2)$

$d = \frac{|1(1) + 2(-1) - 4|}{\sqrt{(1)^2 + (-1)^2}}$

$= \frac{5}{\sqrt{2}}$

$= 3.535..$

now $\sqrt{18} = 4.24..$

since $3.535 < 4.24..$ it cuts the circle at 2 places

Bold answer $\rightarrow$ 2 times 1 mark

- Using Δ function with working
- Proof diagrammatically showing centre & correct evidence $\rightarrow$

3. $\sin 105^\circ = \sin(60+45)$
 $[3] = \sin 60 \cos 45 + \cos 60 \sin 45$
 $= \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} + \frac{1}{2} \cdot \frac{1}{\sqrt{2}}$
 $= \frac{\sqrt{3}+1}{2\sqrt{2}}$ or $\left(\frac{\sqrt{6}+\sqrt{2}}{4}\right)$

4. $y = \frac{x}{3} - 2 \rightarrow m_1 = \frac{1}{3}$
 $[3] 2x + y + 1 = 0$
 $y = -2x - 1 \rightarrow m_2 = -2$

$\tan \alpha = \frac{\frac{1}{3} + 2}{1 + (\frac{1}{3})(-2)}$

$= \frac{2\frac{1}{3}}{\frac{1}{3}}$

$= 7$

$\alpha = 82^\circ$

5 (ii) $\frac{\sin 2x}{1 - \cos 2x} = \cot x$

[4] LHS $= \frac{2 \sin x \cos x}{1 - (1 - 2 \sin^2 x)}$

$= \frac{2 \sin x \cos x}{2 \sin^2 x}$

$= \frac{\cos x}{\sin x} = \cot x = \text{RHS.}$

(ii) $\therefore \cot 15^\circ = \frac{\sin 30^\circ}{1 - \cos 30^\circ}$
 $= \frac{\frac{1}{2}}{1 - \frac{\sqrt{3}}{2}}$
 $= \frac{\frac{1}{2}}{\frac{2 - \sqrt{3}}{2}}$
 $= \frac{1}{2 - \sqrt{3}}$ (or $2 + \sqrt{3}$)

6. $\frac{d}{dx} \left(\frac{(2x-1)^3}{\sqrt{2x-1}} \right) = \frac{d}{dx} (2x-1)^{\frac{5}{2}}$
 $[2] = \frac{5}{2} (2x-1)^{\frac{3}{2}} \times 2$
 $= 5 \sqrt{(2x-1)^3}$

If they use quotient rule 1 mark for correct substitution then 1 mark for correct simplification.

11. $y' = \frac{6(2x-1)^2 \sqrt{2x-1} - (2x-1)^3}{(\sqrt{2x-1})^2}$
 $= \frac{6(2x-1)^2(2x-1) - (2x-1)^3}{(2x-1)\sqrt{2x-1}}$

$= \frac{5(2x-1)^3}{(2x-1)^{\frac{3}{2}}}$
 $= 5(2x-1)^{\frac{3}{2}}$
 $= 5 \sqrt{(2x-1)^3}$

7. $2 \sin \theta - \cos \theta = 1$ using t method.

[4] $\frac{4t}{1+t^2} - \frac{1-t^2}{1+t^2} = 1$

$\therefore 4t - 1 + t^2 = 1 + t^2$

$4t = 2$

$t = \frac{1}{2}$

$\therefore \tan \frac{\theta}{2} = \frac{1}{2}$

$\frac{\theta}{2} = 26^\circ 34', 206^\circ 34'$

$\theta = 53^\circ 08'$

test $\theta = 180^\circ$

$2 \sin 180 - \cos 180 = 1$

$0 - (-1) = 1$

True

$\therefore$ Solns are $53^\circ 08', 180^\circ$

8. $A \sin(\theta + \alpha) = \sqrt{3} \sin \theta + \cos \theta$
 $A \sin \theta \cos \alpha + A \cos \theta \sin \alpha = \text{RHS}$

[4] $A \sin \alpha = 1$ $A \cos \alpha = \sqrt{3}$
 $\therefore \tan \alpha = \frac{1}{\sqrt{3}} \rightarrow \alpha = 30^\circ$

$A = \sqrt{(\sqrt{3})^2 + 1^2} \rightarrow A = 2$

$\therefore 2 \sin(\theta + 30^\circ) = \sqrt{3} \sin \theta + \cos \theta$

(ii) solve $2 \sin(\theta + 30^\circ) = 1$

$\sin(\theta + 30^\circ) = \frac{1}{2}$

$\theta + 30^\circ = 30^\circ, 150^\circ$

$\therefore \theta = 0, 120^\circ, 360^\circ$

9. $f(x) = \frac{2}{x^2}$

$f'(x) = \lim_{h \rightarrow 0} \frac{2}{(x+h)^2} - \frac{2}{x^2}$

[3] $= \lim_{h \rightarrow 0} \frac{2x^2 - 2(x^2 + 2xh + h^2)}{x^2(x+h)^2}$

$= \lim_{h \rightarrow 0} \frac{2x^2 - 2x^2 - 4xh - 2h^2}{x^2(x+h)^2}$

$= \lim_{h \rightarrow 0} \frac{-4x - 2h}{x^2(x+h)^2}$

(NB must carry lim throughout working)

$= \frac{-4x - 0}{x^2(x+0)^2}$
 $= \frac{-4x}{x^4}$
 $= \frac{-4}{x^3}$

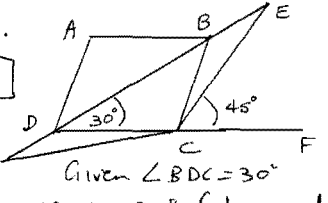
10. $2 \sin 2x = \cos x$
 $0 \leq x \leq 360^\circ$

$4 \sin x \cos x - \cos x = 0$
 $\cos x (4 \sin x - 1) = 0$

$\cos x = 0$ $\sin x = \frac{1}{4}$

$x = 90^\circ, 270^\circ$ $x = 14^\circ 29', 165^\circ 31'$

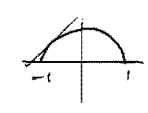
$x = 14^\circ 29', 165^\circ 31', 90^\circ, 270^\circ$

11. 

Given $\angle BDC = 30^\circ$ (1)
 $\angle ADB = 30^\circ$ (diagonals of a rhombus bisect $\angle$ at corner)
 $\therefore \angle ADC = 60^\circ$
 $AD \parallel BC$ (Opposite sides of a rhombus)
 $\angle ADC = \angle BCF = 60^\circ$ (1)
 $\angle ECF = 45^\circ$ (Given)
 $\therefore \angle BCE = 15^\circ$ (Subtraction)
 $\angle ADB = \angle DBC$ (Alternate $\angle$ s $AD \parallel BC$)
 $= 30^\circ$
 $\angle EBC + 30^\circ = 150^\circ$ (Straight $\angle$) (1)
 $\therefore \angle EBC = 150^\circ$
 $\angle BEC = 180 - (150 + 15)$
 $\therefore \angle BEC = 15^\circ$ (1)
 $\therefore \angle BCE = \angle BEC$
 $\triangle BEC$ is isosceles
 (NB $\rightarrow$ other methods award marks accordingly)

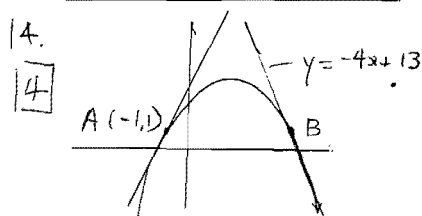
Let $BD = x$
 $\therefore DH = 2x$ (Given $DH = 2BD$)
 Construct AC . let it intersect BD at H . let $AC = h$
 $\therefore$ Area Rhombus $= \frac{1}{2} AC \cdot BD$
 $= \frac{1}{2} \cdot x \cdot h$ (1)
 Area of $\triangle = \frac{1}{2} \cdot 3x \cdot \frac{1}{2} h$ for either
 $= \frac{3xh}{4}$
 $\therefore$ Ratio $\frac{3xh}{4} : \frac{1}{2} xh$
 If they calculate sides & heights $= \frac{3}{4} : \frac{1}{2}$
 if correct award 2 = 3:2. (1)

12. $y = \sqrt{1-x^2} = (1-x^2)^{\frac{1}{2}}$
 $y' = \frac{1}{2} (1-x^2)^{-\frac{1}{2}} \cdot -2x$
 $= \frac{-x}{\sqrt{1-x^2}}$ (1)
 Normal at $135^\circ \Rightarrow$ tangent at 45°
 $\therefore$ grad. of tangent $= 1$
 hence $\frac{-x}{\sqrt{1-x^2}} = 1$ (1)
 $-x = \sqrt{1-x^2}$

$x^2 = 1-x^2$
 $2x^2 = 1$
 $x^2 = \frac{1}{2}$ (1) 
 $x = \pm \frac{1}{\sqrt{2}}$
 but since graph is a semi circle
 $x = -\frac{1}{\sqrt{2}}$ is the only point where this is true
 $y = \sqrt{1 - (-\frac{1}{\sqrt{2}})^2}$
 $= \sqrt{1 - \frac{1}{2}}$
 $= \frac{1}{\sqrt{2}}$ (1)
 $\therefore$ Point is $(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

13. $\frac{6^{\sin x} + 3^{\sin x}}{2^{1+\sin x} + 2} = \frac{3\sqrt{3}}{2}$
 $\frac{6^a + 3^a}{2^{a+1} + 2} = \frac{3\sqrt{3}}{2}$
 let $a = \sin x$
 $\frac{6^a + 3^a}{2^{a+1} + 2} = \frac{3\sqrt{3}}{2}$
 $\frac{3^a(2^a + 1)}{2(2^a + 1)} = \frac{3\sqrt{3}}{2}$ (1)

$\therefore 3^a = 3^{\frac{1}{2}}$
 $\therefore \sin x = \frac{1}{\sqrt{3}}$ (1)
 $x = 19^\circ 26', 160^\circ 32'$ (1)

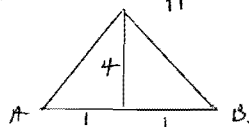


$y = ax^2 + bx + c$
 $(-1, 1)$ satisfies the eq'n
 $a - b + c = 1$ - (1)
 $y' = 2ax + b$ give when
 $x = -1$ $y' = 4$
 $\therefore 4 = -2a + b$ - (2)

Eq'n of tangent at A
 $y - 1 = 4(x + 1)$
 $y = 4x + 5$

Find pt. of intersection
 of $y = 4x + 5$ & $y = -4x + 13$

$4x + 5 = -4x + 13$
 $8x = 8$
 $x = 1 \rightarrow$ this must
 be the axis of sym. of the
 parabola since gradients at
 A & B are opposites



$\therefore$ when $x = 1$ $y' = 0$
 $0 = 2a + b$ - (3)
 solve (2) & (3) simult'ly
 $(2) - (3)$ $4 = 2b \rightarrow b = 2$
 $\therefore a = -1$

sub into (1)
 $-1 - 2 + c = 1$
 $c = 4$

$\therefore a = -1, b = 2$ and $c = 4$

Question 14

Award 1 mark for a
 correct equation in a, b and c
 Award 2 marks for
 2 or 3 correct eqns in
 a, b and c

Award 3 marks if they
 work significantly towards
 a sol'n.

Award 4 marks - if they
 then solve for a, b & c.